My own test rejected my best method.

It couldn’t see a win that size.

I built an open-source benchmark: seven ways to build a portfolio, 18 US ETFs, 168 months, after trading costs, one rule fixed before the run. Then I re-read the result with the test the design was built for, and ran it on a market I had never seen.



The test’s 80% detection threshold, in net excess Sharpe: return above cash per unit of risk, after trading costs. An improvement at the bar would usually be missed.

What this is

A benchmark for portfolio methods: seven constructions, one decision rule, and a test suite that checks the evaluation itself. Not a trading strategy, and not investment advice.

The edge didn’t survive a new market. The lesson did.

Portfolio construction is an estimation problem.

The standard method estimates average returns and correlations from history, then picks the portfolio with the best trade-off between return and risk. The estimates are noisy: mean–variance needs 189 parameters, and the estimation window gives it 60 months of data.

189 parameters

60 months

Regularization — shrinking noisy estimates — is the standard response. Whether it helps is an empirical question, not a mathematical one.

The question: do the "robust" versions — a shrinkage covariance, a Black–Litterman mean — beat a plain baseline out of sample, after costs, once you account for having tried more than one method?

A portfolio is a set of assumptions. The market does not have to share them.

A benchmark, not a backtest.

A backtest is a model of a market. A benchmark tests whether the model's edge survives data it has not seen — and whether the test could see the edge at all.

The engine	A walk-forward test: re-estimate each month on the previous 60 months, then test the next. 168 out-of-sample months in 14 folds.
The universe	18 US ETFs across equity, bonds, commodities and real estate; eligibility rules applied at each date.
The methods	Five standard baselines plus two variants that change one assumption in the standard mean-variance model.
The apparatus	A decision rule fixed before the run, a leakage test suite, a numerical check, and a reproduction check.

1176 optimizer solves

0 numerical rejections

Not a live system, a forecast, or a claim of alpha.

Then I fixed the rule that would decide it.

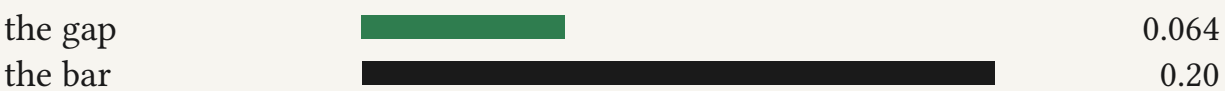
First on the metric. Failed on the decision.

0.8538

net excess Sharpe at 10 bps one-way costs — the highest of the seven

The rule, fixed before the run: beat the best baseline by 0.20, clear the significance test, and pass six robustness checks.

Gap over the best baseline	+0.064
Materiality bar	0.20
Original test (probability edge is real)	0.587 of 0.95
Checks failed	3 of 8



A ranking and a verdict are different objects.

The number that failed said more about the test than the method.

The bar was 0.20. The test needed 0.72.

A test has a detection threshold: the smallest true effect it can reliably notice. This one needed about 0.72 for an 80% chance; the bar was 0.20. A real 0.20 improvement would usually have been missed.



The candidate scored 0.587 against a 0.95 bar. A failing score is not evidence of absence: at the bar, the test would usually have failed.

What went wrong

The test compared the candidate's Sharpe with the baseline's observed Sharpe. The design was built for a paired comparison.

A test can be rigorous and still answer the wrong question.

The same data, the right test.

0.917

probability M2 beats B3 on a paired resampling test

Sharpe gap	+0.064
90% range	−0.011 to 0.147
p-value	0.083

The two strategies hold almost the same portfolio (ρ = 0.986), so a month-by-month comparison is far more sensitive than comparing each to zero.

NOT MATERIAL — 0.147 < 0.20.

The 90% upper bound is below the advantage the study was designed to detect.

A post-hoc cap sweep did not rescue shrinkage; the gap stays at or below zero as the position cap loosens.

The result is a bound, not a shrug.

A new market. A different winner.

18 country ETFs · 2005–2026 · 252 months · 21 folds. The best baseline changed from B3 to B1.

Method	Net Sharpe
B1	0.3527
B2	0.3423
M2	0.3193
B5	0.2919
B3	0.2797
B4	0.2715
M1	0.2710

B1 equal weight · B2 risk parity · B3 sample MVO · B4 minimum variance · B5 turnover-penalised · M1 shrinkage · M2 Black–Litterman.

M2 vs B1: −0.033 [−0.125, 0.075]. **M1 vs B1:** −0.082 [−0.175, 0.029]. Both upper bounds are below 0.20 0 of 5 cost levels; about 3.0× the turnover.

The benchmark lesson replicated. The treatment advantage did not.

The edge depended on the market. The lesson did not.

The benchmark is the binding constraint.

First sample: the top number failed the rule — +0.064 against 0.20.

Second sample: equal weight beat both treatments.

The correction: the same data, read with the test the design was built for.

The limit: every number here belongs to its sample, not to markets in general.

The apparatus: a pre-registered rule, a power audit, and a failed replication on the record.

What I would carry into the next study

— Fix the decision rule before the data.

— Audit the test's power, not only the strategy's return.

— Replicate on a sample that did not inform the design.

Code, paper and the reproduction check: github.com/kira-ml/robustfolio